MBA Semester-IV

Research Project – Interim Report

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“Optimizing ROI on Digital Marketing Campaigns: A data-driven approach on AeroReach Insights "

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Submitted by:

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Declaration

I, Astha Kumari Jha, hereby declare that the Research Project Report titled “Optimizing ROI on Digital Marketing Campaigns: A data-driven approach on AeroReach Insights” has been prepared by me under the guidance of the Hrushikesha Shastry. I declare that this Project work is towards the partial fulfillment of the University Regulations for the award of the degree of Master of Business Administration by Jain University, Bengaluru. I have undergone a project for a period of eight weeks. I further declare that this Project is based on the original study undertaken by me and has not been submitted for the award of any degree/diploma from any other University / Institution.

Place: Bangalore

Date: 29.06.2026

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Objectives of the Study

Defining Problem Statement

Optimizing ROI on Digital Marketing Campaigns: A data-driven approach on AeroReach Insights:

This project aims to develop a predictive and analytical framework that identifies which customer engagement behaviors—such as social media interactions, website visits, and content consumption—are most strongly associated with airline ticket purchases. The model will enable airlines to optimize their digital marketing strategies, prioritize high-performing platforms and content types, and thereby maximize ROI by increasing conversion rates while reducing wasted marketing expenditure.

Figure 1.1: Aviation Digital Marketing ROI Problem Map

Input Challenge	Business Impact	Analytics Solution
Scattered engagement data across travel pages, company pages, and social interactions	Difficulty identifying which users are likely to purchase	Feature engineering and predictive scoring
High campaign spend on broad audiences	Budget wastage on low-intent users	Targeted segmentation using K-Means clusters
Low visibility into conversion drivers	Weak ROI measurement and optimization	XGBoost feature importance and model evaluation

Need of the Study/Project

In the highly competitive aviation industry, digital marketing has become a pivotal channel for customer acquisition and retention. However, many airlines struggle to quantify the direct impact of their digital content on ticket purchase decisions. Traditional marketing strategies often rely on broad metrics, which may not accurately reflect customer behavior or preferences.

This study addresses this gap by developing a predictive model that analyzes how various forms of customer engagement—such as social media interactions, email campaigns, and website visits—affect customers' likelihood to purchase tickets. By integrating data science techniques with marketing analytics, the model aims to provide actionable insights that can guide airlines in optimizing their digital marketing strategies.

By bridging the gap between content engagement and ticket purchase behavior, this research aims to empower airlines to make more informed, data-driven marketing decisions, ultimately enhancing customer acquisition and retention strategies.

This project is necessary because:

- **Marketing Spend is Increasing, but ROI is Not Always Measurable:** Airlines invest heavily in social media and digital ads, yet struggle to quantify the effectiveness of different campaigns.

- **Need to Maximize Conversions While Minimizing Wastage:** In a post-pandemic world with fluctuating travel demand, optimizing every marketing dollar is crucial.
- **Content Engagement ≠ Conversion:** High engagement does not always lead to purchase. Understanding the real predictors of conversions is vital for informed decision-making.
- **Lack of Industry-Specific Models:** Most existing digital marketing analytics frameworks are generic. This study offers a domain-specific approach tailored to the aviation sector.
- **Supports Strategic Planning:** The insights generated will enable AeroReach Insights to advise airlines more effectively, targeting the right audience with the right content at the right time.

Scope of the Study

Business Opportunities:

Enhanced Marketing ROI: By identifying high-performing content and platforms, airlines can allocate resources more effectively, reducing wasted spend and maximizing return on investment.

Personalized Customer Engagement: Understanding the types of content that resonate with different customer segments allows for tailored marketing approaches, leading to improved customer satisfaction and loyalty.

Data-Driven Decision Making: The insights derived from the predictive model enable marketing teams to make informed decisions about content creation, distribution channels, and campaign timing, aligning marketing strategies with customer preferences.

Competitive Advantage: Airlines that leverage data science to optimize their marketing strategies can differentiate themselves in a crowded market, attracting more customers and increasing market share.

Revenue Optimization: By understanding the impact of content engagement on ticket purchases, airlines can implement strategies to increase conversions, leading to higher revenue per customer.

Social Opportunities:

Improved Customer Experience: Personalized content and targeted marketing efforts enhance the overall customer experience, fostering positive relationships between airlines and passengers.

Community Engagement: By analyzing social media interactions, airlines can engage with customers in meaningful ways, building a sense of community and brand loyalty.

Social Responsibility: Understanding customer preferences allows airlines to promote sustainable travel options and social initiatives, aligning with the growing consumer demand for corporate social responsibility.

Cultural Sensitivity: Tailoring content to resonate with diverse cultural backgrounds ensures that marketing efforts are inclusive and respectful, fostering a positive brand image.

Public Health and Safety Communication: During times of crisis, such as the COVID-19 pandemic, effective communication through digital content can disseminate important health and safety information to the public.

Data Collection Method

Time Aspect

Features like:

1. **Yearly_avg_view_on_travel_page**
2. **yearly_avg_Outstation_checkins**
3. **Yearly_avg_comment_on_travel_page**
4. **montly_avg_comment_on_company_page**
5. **week_since_last_outstation_checkin**
6. **Daily_Avg_mins_spend_on_traveling_page**

- **Indicate temporal tracking —** the data appears to be collected periodically over time, likely across monthly/yearly intervals.

Frequency:

- **Data is aggregated per user per year or per month,** based on feature names.
- **It's a snapshot dataset,** where each row is a summary of a user's engagement.

Methodology:

Data is likely collected from:

- **App/website activity logs** (e.g., views, likes, time spent)
- **User profiles** (e.g., device, family members)
- **Engagement records** (comments, check-ins)
- **Marketing touchpoints** (e.g., following company page)
- Could be collected via tracking scripts, user-provided data, and platform metadata

Figure 2.1: Data Collection and Preparation Flow

Website / App Logs	Data Cleaning	Feature Engineering	Model-Ready Dataset
→	→	→	

Understanding of Attributes

Dataset Dimensions

The dataset contains 11,760 rows (users) and 17 columns (features).

Category	Description	Type
UserID	Unique user identifier	Integer
Taken_product	Whether the user purchased a ticket (Yes/No) → Target Variable	Categorical
Yearly_avg_view_on_travel_page	Average yearly page views	Float
preferred_device	Device used to engage (e.g., iOS, Android)	Categorical
total_likes_on_outstation_checkin_given	Total likes user gave to others' check-ins	Float
yearly_avg_Outstation_checkins	Avg yearly outstation check-ins	Numeric
member_in_family	Family size	Numeric
preferred_location_type	Location type preference (e.g., Financial, Medical)	Categorical
Yearly_avg_comment_on_travel_page	Average yearly comments made	Float
total_likes_on_outofstation_checkin_received	Total likes received	Integer
week_since_last_outstation_checkin	Time since last check-in	Integer
following_company_page	Is user following airline's page? Yes/No	
montly_avg_comment_on_company_page	Avg comments on company page	Integer
working_flag	Employment status (Yes/No)	Categorical
travelling_network_rating	Peer rating or network influence	Integer (1–5)
Adult_flag	Age group indicator	Integer (1=Adult, 0=Minor)
Daily_Avg_mins_spend_on_traveling_page	Engagement duration	Integer

Missing Values Overview (out of 11,760 rows)

Column	Missing Values
Yearly_avg_view_on_travel_page	581
preferred_device	53
total_likes_on_outstation_checkin_given	381
yearly_avg_Outstation_checkins	75

Column	Missing Values
preferred_location_type	31
Yearly_avg_comment_on_travel_page	206
following_company_page	103

Data Analysis Tools

Exploratory Data Analysis

Univariate Analysis

Univariate analysis focuses on examining one variable at a time to understand its distribution, central tendency (like mean and median), and spread (range, standard deviation). In the context of this project, we use univariate analysis to explore user engagement behaviors that may influence ticket purchase decisions.

Why Univariate Analysis Matters

This analysis helps us:

- Understand which user behaviors are common vs. rare
- Detect data patterns, outliers, and skewness
- Identify features that might be important predictors in modeling ticket purchases

By visualizing these attributes through histograms and boxplots, we gain clearer insights into how individual engagement metrics behave, laying the foundation for deeper analyses like bivariate analysis and predictive modeling.

Figure 3.1: Engagement Level Distribution of Users

User Segment	Approx. Share
Passive Users	65%
Moderate Users	25%
Highly Engaged Users	10%

The chart indicates that most users are passive viewers, while a smaller but strategically important group shows high engagement. This supports the need for focused campaign targeting rather than broad marketing spend.

Key Observations:

- **Daily Average Minutes Spent on Travel Pages:** Most users spend less than 10 minutes daily browsing travel-related content. The data is right-skewed, meaning a few users spend significantly more time. These high-engagement users may have a higher likelihood of converting (purchasing).

- **Yearly Average Views on Travel Pages:** A wide range of values observed, from under 100 to over 500 views per year. This shows different levels of user interest; higher averages indicate consistent engagement.
- **Total Likes Given on Outstation Check-ins:** Some users show extremely high like activity, indicating strong social interaction. The data includes outliers, which are values significantly higher than most others.
- **Yearly Average Comments on Travel Pages:** The majority of users make fewer than 100 comments per year, but some comment heavily. Commenting behavior reflects deeper user involvement and potential brand loyalty.
- **Total Likes Received on Out-of-Station Check-ins:** Users who receive more likes may have a wider social influence, possibly affecting others' travel decisions.
- **Week Since Last Outstation Check-in:** This variable tells us how recent a user's last trip was. Recent travelers (fewer weeks since last check-in) are possibly more ready to travel again.
- **Travel Network Rating:** A discrete rating scale from 1 to 5. Most users rate their travel network as 3 or 4, showing general satisfaction.
- **Monthly Average Comments on Company Pages:** Varies moderately across users. Frequent interaction on company pages may be linked to brand engagement.

Bivariate Analysis:

Bivariate analysis involves examining the relationship between two variables at a time. In this project, we explore how different engagement features (like time spent, likes, comments) relate to the target variable: whether or not a user has purchased a product (Taken_product = Yes/No).

This analysis helps uncover dependencies, correlations, and influencing factors behind customer behavior in digital marketing campaigns.

Why Bivariate Analysis Is Important

- Helps identify which features have strong relationships with the outcome (purchase or not)
- Useful for feature selection in predictive modeling
- Gives direction for business strategy — e.g., targeting users with high engagement scores

This analysis provides insights into which user behaviors matter most and should be prioritized in digital campaign strategies to maximize ROI.

Key Relationships Observed:

- **Daily Average Minutes Spent on Travel Page:** Users who purchased (Yes) spent more time on travel pages compared to non-purchasers. This shows a positive relationship between time spent and likelihood of purchase.

- **Yearly Average Views on Travel Page:** Purchasers had a higher number of yearly views, suggesting consistent interest in travel-related content increases chances of conversion.
- **Total Likes on Outstation Check-ins Given:** Purchasers tend to give more likes — indicating higher social engagement.
- **Yearly Average Comments on Travel Page:** Users who purchased often left more comments, showing that active content interaction correlates with buying behavior.
- **Total Likes Received on Outstation Check-ins:** Purchasers received slightly more likes — possibly due to higher visibility or social influence.
- **Week Since Last Outstation Check-in:** Users who made purchases had fewer weeks since their last trip, meaning recent travelers are more likely to buy again.
- **Monthly Average Comments on Company Pages:** Purchasers engaged more with company content — this shows brand engagement may drive conversions.

Data Cleaning

Variables removed during data cleaning:

- Have very low variability (same or similar values across rows)
- Are irrelevant to our objective (e.g., identifiers)
- Contain inconsistent or unstructured data that can't be used in analysis

Missing Value Treatment (Post-Cleaning)

Column Name	Missing Values	Suggested Action
Yearly_avg_view_on_travel_page	581	Impute with median
total_likes_on_outstation_checkin_given	381	Impute with median
Yearly_avg_comment_on_travel_page	206	Impute with median
following_company_page	103	Impute with mode (categorical)
yearly_avg_Outstation_checkins	76	Impute with median
preferred_device	53	Impute with mode (categorical)
preferred_location_type	31	Impute with mode (categorical)

Outlier Treatment:

Outlier treatment is completed using the IQR (Interquartile Range) method. This technique is used to cap extreme values that may skew the analysis or model results.

Why It Matters in Project:

Outliers like extremely high “likes given” or “time spent” could:

- Distort model training, leading to biased predictions
- Cause incorrect insights in data visualizations
- Affect ROI optimization logic, especially when working with budget-related features

Variable Transformation:

Some features like “likes”, “views”, and “time spent” had large numeric ranges or skewed distributions. Scaling them:

- Makes them comparable in magnitude
- Improves performance of machine learning models like logistic regression or KNN
- Helps visualization clarity by removing disproportionate ranges

Business Insights from Exploratory Data Analysis (EDA)

7. **High Engagement = Higher Purchase Probability:** Users who spent more time on travel pages, gave more likes, and posted more comments showed a higher likelihood of purchasing a ticket. Insight: Focus digital campaigns on users with high engagement scores to maximize ROI.
8. **Device Preference Matters:** While not yet deeply analyzed in visuals, prior studies show device usage (e.g., Android, iOS) influences content interaction. Insight: Tailor ad formats and creatives based on user device — mobile-first for highly engaged users.
9. **Recency Drives Action:** The more recent a user's travel check-in, the higher their recency score — and higher chance of conversion. Insight: Launch retargeting campaigns within 2-4 weeks of a user's last travel-related activity.
10. **Social Influence = Brand Amplification:** Users who received more likes on their check-ins and had higher travel network ratings tend to engage and purchase more. Insight: Use these users for influencer-like targeting or referral campaigns.
11. **Brand Engagement Correlates with Purchase:** Users who followed company pages and regularly commented on brand content had higher brand engagement scores and were more likely to purchase. Insight: Encourage brand interactions — e.g., polls, Q&A sessions — to build loyalty and increase conversions.
12. **Unbalanced Interest Distribution:** Majority of users are passively engaged (less than 10 minutes daily). A smaller group is highly active, driving most of the conversion. Insight: Use clustering or segmentation to identify and exclusively target high-potential users.
13. **Marketing Spend Optimization Potential:** Although spend wasn't explicitly analyzed, combining conversion rates with engagement data can help in redirecting spend from low-performing audiences and focusing on high ROI platforms or content types.

Strategic Recommendations:

Recommendation	Business Value
Retarget recent travelers	Higher conversion probability
Promote brand interaction content	Builds trust and repeat engagement
Leverage socially influential users	Organic reach + peer impact
Segment users by engagement score	Personalized and efficient ad delivery
Optimize for high-engagement devices	Platform-specific ROI boost

Why Class Imbalance Matters in Business

In the context of digital marketing ROI optimization, correctly identifying users who are likely to purchase is crucial. If the model simply predicts “No” for everyone, it would:

- Appear accurate (~84%), but
- Fail to capture the small but valuable customer segment that actually converts — the target audience for marketing spend.

Technique	Description
Resampling (Over/Under)	Increase the number of "Yes" cases (oversampling) or reduce "No" cases
SMOTE (Synthetic Minority Over-sampling Technique)	Generates synthetic examples of the minority class to balance the dataset
Class Weight Adjustment	Use model parameters to give more importance to "Yes" class
Precision-Recall Metrics	Use metrics like F1-score, recall, or AUC instead of just accuracy

Business Context Recommendation:

- Use oversampling or SMOTE to boost the representation of purchasers.
- Focus analysis and marketing spend on the rare group of users who do convert.
- Implement targeted messaging or personalized ads for high-engagement users detected through modeling — even if they're few.

Business Insights Using Clustering

To uncover patterns within user engagement behavior, K-Means clustering was applied using four engineered features:

- Engagement Score

- Social Influence Score
- Recency Score
- Brand Engagement Score

These metrics were selected as they represent key dimensions of user interaction with travel-related content and brand presence.

Figure 4.1: K-Means Cluster Profile Comparison

Cluster	Engagement	Recency	Brand Score	Priority
Cluster 0: High-Value Converters	High	High	High	Highest
Cluster 1: Moderately Engaged Observers	Medium	Medium	Medium	Nurture
Cluster 2: Passive or Low-Potential Users	Low	Low	Low	Low Spend

Based on the elbow method, the optimal number of clusters was determined to be 3.

Each cluster revealed distinct behavioral segments that offer actionable marketing opportunities:

Cluster 0 – High-Value Converters

- **Profile:** Highly engaged, socially influential, and recently active users with strong brand interactions.
- **Business Insight:** This group represents the most promising segment for targeted conversion campaigns. Users in this cluster are likely to respond positively to personalized offers, referral programs, and exclusive travel deals.
- **Recommendation:** Prioritize marketing spend here to maximize ROI.

Cluster 1 – Moderately Engaged Observers

- **Profile:** Users with average engagement levels and low social visibility. Their travel activity is less recent, and brand interaction is moderate.
- **Business Insight:** This segment exhibits latent interest and can be nurtured through remarketing strategies or loyalty-building campaigns.
- **Recommendation:** Use email re-engagement or value-driven content to move them toward purchase intent.

Cluster 2 – Passive or Low-Potential Users

- **Profile:** Users with minimal engagement, low social influence, and long inactivity since last travel interaction.
- **Business Insight:** These users are less likely to convert in the short term and may not provide significant ROI on direct marketing investment.

- **Recommendation:** Focus on awareness campaigns or limit ad spend for this group. Alternatively, explore survey-based approaches to better understand their disengagement.

Other Business Insights

Beyond the direct relationships and clustering results, several additional patterns emerged during the exploratory analysis. These insights provide strategic guidance for campaign optimization, content development, and marketing budget allocation.

1. Behavioral Signals Predict Purchase Intent

Multiple features, such as average daily time spent, comment activity, and content interaction, showed strong correlation with purchase behavior. Users who consistently engage across multiple touchpoints are more likely to convert.

- **Business Action:** Develop composite engagement-based scoring models to identify high-intent users in real time.

2. Brand Touchpoints Drive Conversions

Users who frequently interact with the brand's company page — by commenting or following — tend to show higher purchase probability. This suggests that brand familiarity builds trust, which leads to conversions.

- **Business Action:** Increase interactive content (polls, Q&A, behind-the-scenes posts) to boost engagement on brand-owned pages.

3. Recency is a Critical Purchase Trigger

Users who recently engaged in travel-related activity (e.g., check-ins) are more likely to purchase again soon. This behavioral recency is a strong predictor of re-conversion.

- **Business Action:** Implement dynamic remarketing strategies that target users within 2-4 weeks of their last travel-related engagement.

4. Not All Engagement is Equal

While likes are abundant, comments and time spent are more indicative of serious interest. These “deep engagement” signals are better predictors of conversions than surface-level actions.

- **Business Action:** Refocus metrics for campaign performance from vanity metrics (likes) to depth metrics (comments, time-on-page).

5. Platform-Specific Optimization Opportunity

Although not deeply covered here, different platforms (e.g., Instagram vs. YouTube) may perform differently across content types. Users may engage more with short-form videos on one platform, while preferring infographics or articles on another.

- **Business Action:** Perform platform-wise engagement mapping to align content formats with channel performance for maximum ROI.

6. Marketing Spend Should Follow Data

By combining conversion data with engagement metrics, AeroReach can reallocate marketing budgets away from low-performing user segments or content types and invest more in high-impact channels and behaviors.

- **Business Action:** Introduce a performance-based budget allocation model tied to predictive scoring.

Strategic Implications

These insights emphasize the importance of:

- Granular behavioral tracking
- Timely user segmentation
- Platform-aware campaign strategies
- Data-driven content personalization

With proper integration into AeroReach Insights' marketing intelligence platform, these findings can lead to significant improvement in conversion rates, reduced wastage, and maximized return on digital advertising investments.

Methodology: Model Building and Interpretation

To achieve the objective of identifying users most likely to purchase tickets based on their engagement behavior, an XGBoost classification model was developed. XGBoost is a gradient boosting algorithm that builds a sequence of decision trees, where each new tree improves upon the errors of the previous trees. This makes it suitable for conversion prediction because it can learn complex behavioral patterns and generate purchase probability scores for campaign targeting.

Figure 5.4: XGBoost Modeling Workflow

Clean Data	Train-Test Split	SMOTE + Weighting	XGBoost	Buyer Probability Score
→	→	→	→	

1. Problem Framing

- **Type of Problem:** Supervised classification
- **Target Variable:** Taken_product (Yes = Purchased, No = Not Purchased)
- **Objective:** Predict the probability of a user purchasing a ticket based on engagement, brand interaction, and social behavior features.

2. Feature Selection

The following features were used as predictors:

Engineered scores:

- engagement_score
- social_influence_score
- brand_engagement_score
- recency_score

Original features:

- Daily_Avg_mins_spend_on_traveling_page
- Yearly_avg_comment_on_travel_page
- week_since_last_outstation_checkin
- travelling_network_rating

These were selected based on EDA insights, correlation with purchase behavior, and business logic.

3. Handling Class Imbalance

As only 16% of users in the dataset had made a purchase, the data was highly imbalanced. To address this, the modeling approach used both data-level and algorithm-level balancing techniques:

- **SMOTE (Synthetic Minority Oversampling Technique)** was used to synthetically generate more samples for the minority class (Yes).
- **The XGBoost scale_pos_weight parameter** was used to assign higher importance to the minority class, helping the model focus on likely buyers rather than over-predicting the majority “No” class.

4. Model Selection and Training

Several models were tested for performance:

- Logistic Regression
- Random Forest
- XGBoost Classifier

The XGBoost classifier was selected due to its:

- Strong predictive performance for structured and tabular customer behavior data
- Ability to capture complex nonlinear relationships between engagement signals and purchase decisions

- Better control over class imbalance through parameters such as scale_pos_weight, along with feature importance interpretation

The dataset was split:

- **Training Set:** 80%
- **Testing Set:** 20%

5. Evaluation Metrics

Given the business importance of identifying likely buyers, emphasis was placed on:

- **Precision:** Correctly identifying actual buyers
- **Recall:** Capturing all potential buyers (sensitivity)
- **F1 Score:** Balanced measure
- **ROC-AUC:** Discrimination capability across thresholds

Model Performance Evaluation

6. XGBoost Results Interpretation

- **Accuracy (post-balancing):** ~97%
- **Precision:** Very high — the model minimizes wasted spend by correctly identifying users who are likely to buy
- **Recall:** Strong — the model captures most actual buyers, reducing missed conversion opportunities
- **AUC Score:** Excellent — indicates strong separation between buyer and non-buyer classes

Confusion Matrix

XGBoost Classifier Confusion Matrix

Actual / Predicted	No (0)	Yes (1)
No (0)	1966	7
Yes (1)	67	318

Figure 5.1: The green cells represent correct predictions, while the orange cells represent errors. The model performs strongly because false positives are very low and most actual buyers are correctly identified.

Performance Metrics

Metric	Value	Interpretation
Accuracy	97.11%	Overall correct predictions

Metric	Value	Interpretation
Precision (Class 1)	97.85%	When model predicts "Yes", it's correct 97.8% of the time
Recall (Class 1)	83.91%	The model detects 82.3% of actual buyers
F1 Score (Class 1)	90.41%	Good balance between precision and recall
ROC-AUC Score	0.996	Excellent at distinguishing between buyer and non-buyer classes

Figure 5.2: XGBoost Performance Metrics

Metric	Value
Accuracy	97.11%
Precision	97.85%
Recall	83.91%
F1 Score	90.41%
ROC-AUC	0.996

7. XGBoost Feature Importance (Business Relevance)

Figure 5.3: Gain-Based Feature Importance

Feature	Rank
engagement_score	1
recency_score	2
brand_engagement_score	3
social_influence_score	4
Feature Name	**Interpretation**
engagement_score	Highest gain contribution — users with stronger browsing, commenting, and liking behavior are more likely to purchase
recency_score	Recent travelers show stronger short-term booking intent and should be prioritized for remarketing
brand_engagement_score	Brand-page interactions improve conversion probability by reflecting trust and familiarity

Feature	Rank
social_influence_score	Socially active users influence travel decisions and can support referral or advocacy campaigns

Figure 6.1: Predictive Marketing Action Funnel

- All Digital Audience Users
- Segmented by Engagement, Recency, Brand Interaction, and Social Influence
- XGBoost Predicts Likely Buyers
- Marketing Team Targets High-Probability Users
- Improved Conversion Rate and Reduced Budget Wastage

Business Utility of the Model

- Enables campaign personalization for high-likelihood users
- Helps reduce spend on unlikely converters
- Empowers AeroReach to build data-driven audience segments
- Supports real-time predictive targeting in digital campaigns

Business Interpretation

- **High Precision (97.85%)** ensures that users flagged as “likely to buy” are highly reliable targets for paid campaigns.
- **Strong Recall (83.91%)** indicates that XGBoost identifies most potential buyers, reducing missed revenue opportunities.
- **The ROC-AUC of 0.996** demonstrates excellent discriminatory power between buyer and non-buyer classes.

Marketing Implication

- Use this model to score and segment users in real-time.
- Focus marketing efforts (e.g., email offers, remarketing ads) on users with high predicted probabilities of purchase.
- Reduce budget wastage on uninterested or low-engagement users.

Conclusion & Future Work

Conclusion

This study successfully demonstrated the application of data science techniques to analyze user engagement and optimize return on investment (ROI) in digital marketing campaigns within the aviation industry.

Using real-world behavioral and interaction data, a predictive model was developed to identify users most likely to purchase airline tickets. Through rigorous Exploratory Data Analysis (EDA), feature engineering, and clustering, valuable business insights were generated, leading to the identification of high-value user segments.

The XGBoost classifier, trained on key engagement and brand interaction metrics, achieved impressive performance:

- **Accuracy:** 97.11%
- **Precision (Yes class):** 97.85%
- **Recall:** 83.91%
- **AUC Score:** 0.996

These results confirm that XGBoost is a strong alternative model for this study. Its ability to handle nonlinear relationships, imbalanced data, and feature importance analysis makes it suitable for identifying potential buyers and supporting ROI-driven campaign decisions for AeroReach Insights.

Future Work

While the current analysis provides a solid foundation, several avenues can be explored to further enhance model performance and business impact:

- **Platform-wise Campaign Analysis** ensures that marketing efforts focus on the most effective channels, optimizing budget allocation.
- **Text Analytics on Comments** brings qualitative insights by understanding audience sentiment — crucial for refining messaging and engagement strategies.
- **Real-time Scoring System** offers dynamic audience segmentation, allowing for immediate, data-driven campaign adjustments.
- **A/B Testing with Model-Based Targeting** provides empirical validation on the effectiveness of AI-driven segmentation, helping refine targeting efficiency.
- **Deeper Demographic Segmentation** enhances personalization by incorporating granular user insights.
- **Time Series Campaign Impact Tracking** assists in spotting trends, helping with forecasting and seasonal strategy refinement.

By continuously iterating and scaling this model, AeroReach Insights can lead the way in data-driven airline marketing, driving conversions more intelligently and cost-effectively.